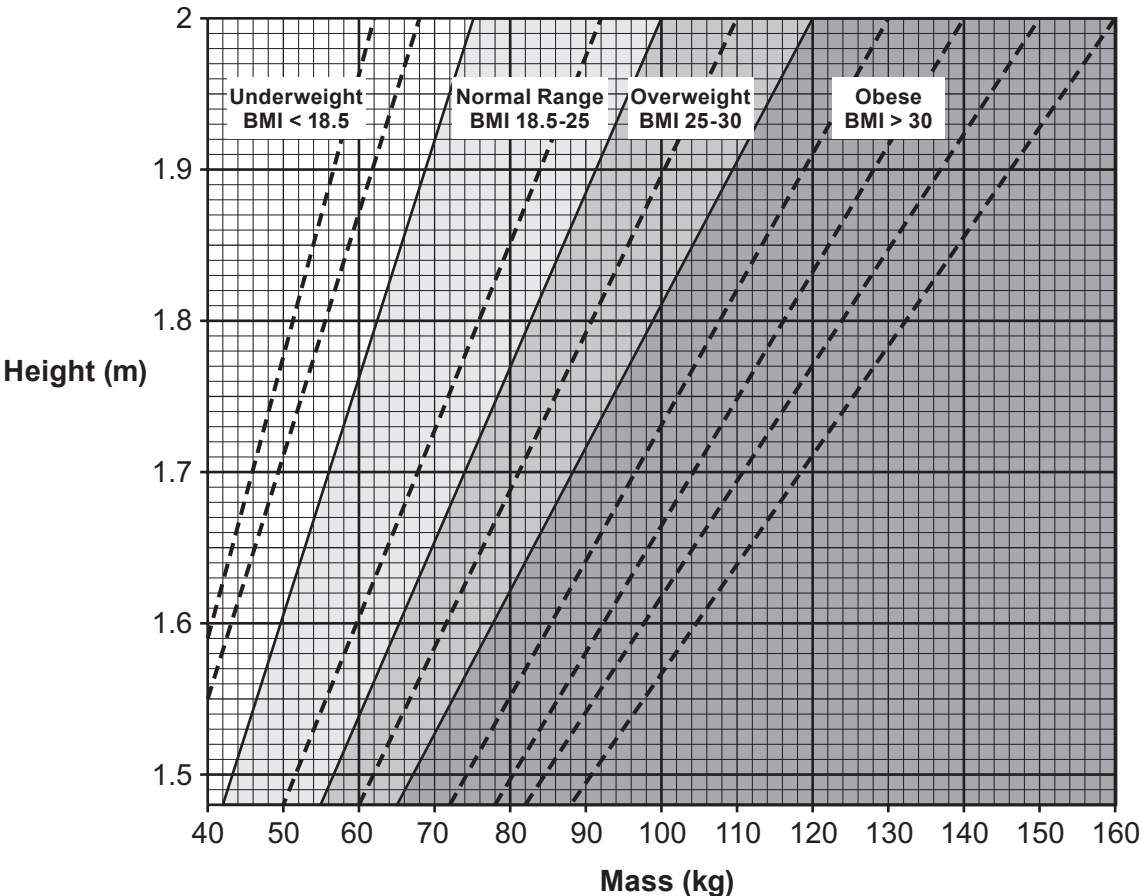


4. The Body Mass Index (BMI) is one way of measuring whether a person is a healthy mass for their height. The result gives an indication of whether the person is a healthy mass, and if not, how overweight or underweight they are.
The BMI is calculated from the mass and height of a person. It is calculated using the following equation:

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

where body mass is measured in kilograms and height in metres.

The chart shows the BMI categories based on the World Health Organisation data. The dashed lines represent subdivisions within a major category. For example, the underweight category is subdivided into severe, moderate and mild.



Source: Calculator.net

- (a) From the chart, state how many subdivisions there are within the obese category. [1]
number of subdivisions in the obese category =



Examiner
only

(b) The table below shows some biometric data for three teachers at a school.

Teacher	Mass (kg)	Height (m)	BMI	BMI category
Iorwerth	82.4	1.56	33.9	obese
Sharon	55.1	1.71	18.8	
Peter	95.3	1.84		

(i) Calculate the BMI for Peter and **write the answer in the table.** [2]
Space for working.

(ii) Use the chart opposite to allocate a major BMI category to Sharon and Peter and **write the answer in the table** above. [1]

(c) Iorwerth wants to reduce his BMI. State **two** ways in which he could alter his diet in order to achieve this. [2]

- I.
- II.

